11 The cumulative frequency table gives information about the time, in minutes, Jane took to go from her home to school each day last term.

Time taken (t minutes)	Cumulative frequency
$0 < t \leqslant 10$	0
$0 < t \leqslant 20$	7
$0 < t \leqslant 30$	20
$0 < t \leqslant 40$	64
$0 < t \leqslant 50$	74
$0 < t \leqslant 60$	80

(a) On the grid opposite, draw a cumulative frequency graph for this information.

(2)

Jane expects that it should take her x minutes to go from her home to school each day.
On 25% of the days last term, Jane took longer than x minutes to go from her home to school.

(b) Use your cumulative frequency graph to find an estimate for the value of x.

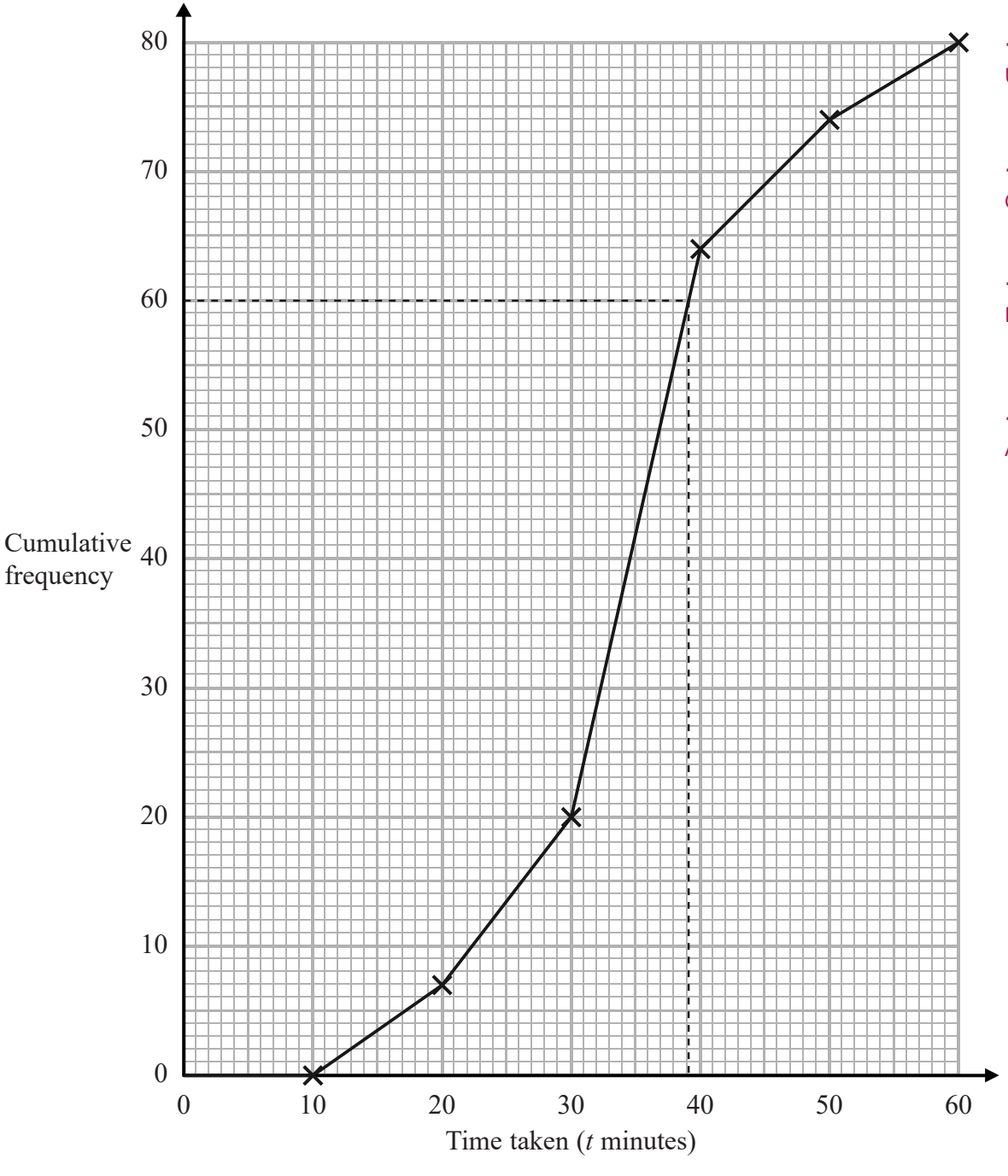
25% of80 = 80 ÷ 4 = 20 days above x.

So80 - 20 = 60 days at or below x.

Read across from CF60, then down: x ≈ 39 minutes.

1 mark
25% of80 =20

(3)



1 mark
Upper endpoints

1 mark
Correct CF graph

1 mark
Read from60

1 mark
About39 min

75% of the days are at or below x (upper quartile).

(Total for Question 11 is 5 marks)